CSCE 4813 - Computer Graphics
Midterm Exam – Spring 2015

Student Name: Solutions.

Student UAID: _____

I pledge that I have neither given nor received unauthorized help on this exam

Instructions: This is closed book exam. Students are allowed one 8.5 x 11 page of notes, but no calculators or other electronic devices. The exam worth a total of 60 points.

1) Geometric Models and Transformations (20 points)

We have discussed rotation, translation and scaling transformations in class. For this question, we will be using the following notation:

- R(theta, axis) denotes a rotation by theta degrees around the specified (x, y, z) axis
- T(dx, dy, dz) denotes a translation by (dx, dy, dz) units
- S(sx, sy, sz) denotes a scaling by (sx, sy, sz) units

a) What is the formula for T(dx, dy, dz)? Show how this formula can be implemented using a 4x4 matrix multiplication.

$$x' = x + dx$$
$$y' = y + dy$$
$$z' = z + dz$$

$$[x' \; y' \; z' \; 1] = \begin{bmatrix} 1 & 0 & 0 & dx \\ 0 & 1 & 0 & dy \\ 0 & 0 & 1 & dz \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

b) What is the formula for S(sx,sy,sz)? Show how this formula can be implemented using a 4x4 matrix multiplication.

$$x' = x \cdot Sx$$
$$y' = y \cdot Sy$$
$$z' = z \cdot Sz$$

$$[x' \; y' \; z' \; 1] = \begin{bmatrix} S_x & 0 & 0 & 0 \\ 0 & S_y & 0 & 0 \\ 0 & 0 & S_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

c) What is the formula R(theta, z-axis)? Show how this formula can be implemented using a 4x4 matrix multiplication.

$$x' = x\cos\theta - y\sin\theta$$
$$y' = x\sin\theta + y\cos\theta$$
$$z' = z$$

$$[x' \; y' \; z' \; 1] = \begin{bmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

d) Sometimes we want to scale an object around its center point (cx,cy,cz) to simplify user interaction or to get different visual effects. What sequence of geometric operations is needed to perform this operation?

$T(-cx, -cy, -cz)$ – move object to origin

$S(sx, sy, sz)$ – scale object

$T(cx, cy, cz)$ – move object to $(cx\ cy\ cz)$

2) Computer Viewing (20 points)

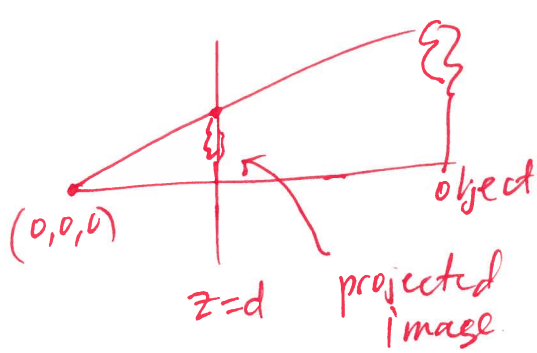
The goal of projection is to transform three-dimensional object coordinates (X, Y, Z) into two-dimensional screen coordinates (xp, yp).

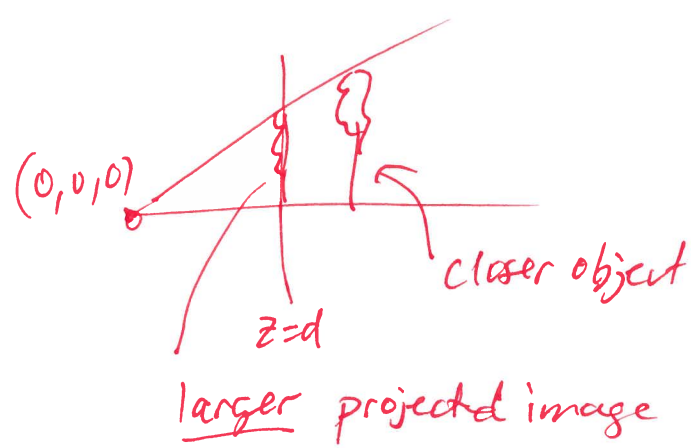
a) For perspective projection, assume the center of projection is at (0, 0, 0) and the projection plane is at Z=d. What are the formulas for calculating (xp, yp) in terms of (X, Y, Z) and d?

$$xp = \frac{X \cdot d}{Z}$$

$$yp = \frac{y \cdot d}{Z}$$

b) Draw a side view diagram to illustrate what happens to the projected image if we move an object closer to the projection plane. Will the image become larger or smaller?



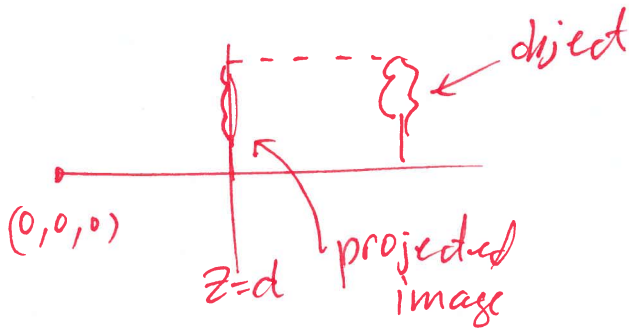


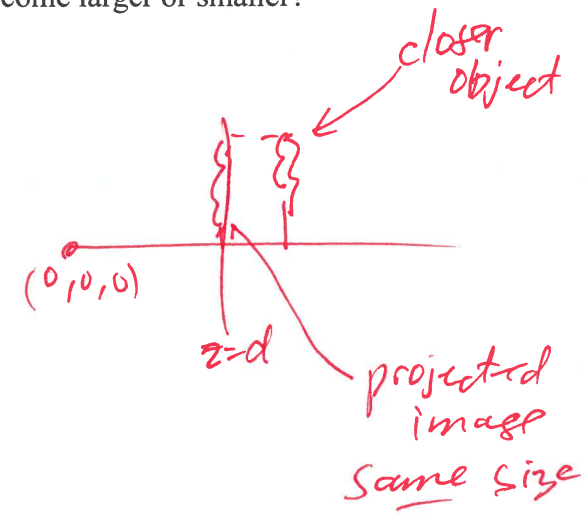
c) For parallel projection, assume projection plane is at Z=d. What are the formulas for calculating (xp, yp) in terms of (X, Y, Z) and d?

$x_p = x$

$y_p = y$

d) Draw a side view diagram to illustrate what happens to the projected image if we move an object closer to the projection plane. Will the image become larger or smaller?





3) Line Drawing (20 points)

Consider the following line drawing code:

```
int Image[20][20];
void draw_line(int x1, int y1, int x2, int y2, int value)
{
   // Calculate step size
   double dx = x2 - x1;
   double dy = y2 - y1;

   // Draw points on line
   double x = x1 + 0.5;
   for (int y = y1; y <= y2; y++)
   {
      Image[y][(int)x] = value;
      x += dx/dy;
   }
}
```

a) What pixels in the image will be filled in when drawing a line from (10,5) to (15,15)?

x	(int) x	y
10.5	10	5
11	11	6
11.5	11	7
12	12	8
12.5	12	9
13	13	10
13.5	13	11
14	14	12
14.5	14	13
15	15	14
15.5	15	15

these are the pixel coordinates

b) What will happen if we try to draw a line with dx < 0?

- the line will be drawn correctly
- the x values will decrease.

c) What will happen if we try to draw a line with dy < 0?

- the line will <u>not</u> be drawn because $y_2 < y_1$ loop executes zero times.

d) What will happen if we try to draw a line with dy = 0?

- program will crash with divide by zero error.